

we refer to β exposure – meaning all tasks directly exposed via tools like ChatGPT or the OpenAI Playground are considered twice as exposed as tasks requiring some complementary innovation.

3.4 Limitations of our methodology

3.4.1 Subjective human judgments

A fundamental limitation of our approach lies in the subjectivity of the labeling. In our study, we employ annotators who are familiar with LLM capabilities. However, this group is not occupationally diverse, potentially leading to biased judgments regarding LLMs’ reliability and effectiveness in performing tasks within unfamiliar occupations. We acknowledge that obtaining high-quality labels for each task in an occupation requires workers engaged in those occupations or, at a minimum, possessing in-depth knowledge of the diverse tasks within those occupations. This represents an important area for future work in validating these results.

3.4.2 Measuring LLMs with GPT-4

Recent research indicates that GPT-4 serves as an effective discriminator, capable of applying intricate taxonomies and responding to changes in wording and emphasis (OpenAI, 2023b). The outcomes of GPT-4 task classification are sensitive to alterations in the rubric’s wording, the prompt’s order and composition, the presence or absence of specific examples in the rubric, the level of detail provided, and the definitions given for key terms. Iterating on the prompt, based on observed outcomes in a small validation set, can enhance the agreement between model outputs and the rubric’s intent. Consequently, there are slight differences between the rubric presented to humans and the one used for GPT-4. This decision was made deliberately to guide the model towards reasonable labels without excessively influencing human annotators. As a result, we use multiple annotation sources, but none should be considered the definitive ground truth relative to the others. In this analysis, we present results from human annotators as our primary results. Further improvement and innovation in crafting effective rubrics for LLM classification remains possible. Still, we observe a high degree of agreement between human ratings and GPT-4 ratings at the occupation level concerning overall exposure to LLM systems (see Table 2, Figure 2).

3.4.3 Additional Weaknesses

- **Validity of task-based framework.** It is unclear to what extent occupations can be entirely broken down into tasks, and whether this approach systematically omits certain categories of skills or tasks that are tacitly required for competent performance of a job. Additionally, tasks can be composed of sub-tasks, some of which are more automatable than others. Some tasks may function as pre-cursor to other tasks, such that the completion of downstream tasks is dependent on precursor tasks. If indeed, the task-based breakdown is not a valid representation of how most work in an occupation is performed, our exposure analysis would largely be invalidated.
- **Lack of expertise and task interpretation.** Human annotators were mostly unaware of the specific occupations mapped to each DWA during the labeling process. This led to unclear logic for aggregating tasks and occupations, as well as some evident discrepancies in labels, demonstrated in Table 1. We experimented with various aggregation methods and discovered that even with a maximum-matching approach (taking the matching human<>model label if one existed), the agreement remained relatively consistent. Ultimately, we collected additional labels for task/occupation pairs where there was significant disagreement.